

Esercizio 1. Let $N \models \mathcal{T}$, ec for $\mathcal{T}^+$, such that $N \models \Phi(a)$. Let $p(x) = pp - tp(a)$ (intended in the new language). Then $\mathcal{T} \cup p(x) \cup \{\exists y \phi(x, y)\}$ is consistent. As a consequence, $N \models \exists y \phi(a, y)$.

Proof. By contradiction, let's say it is inconsistent. Then it is finitely inconsistent. Then $T \models \neg(\exists x \exists y \phi(x, y) \wedge \bigwedge_{k=1, \dots, n} \theta_k(x))$ for some $\theta_1, \dots, \theta_n \in p(x)$. Conjunction of pp formulae is equivalent to a pp formula, so by definition of $\mathcal{T}^+$ we have $\neg(\exists x \Phi(x) \wedge \bigwedge_{k=1, \dots, n} \theta_k(x)) \in \mathcal{T}^+$. Then $N \models p(a) \wedge \Phi(a) \wedge \neg(\exists x \Phi(x) \wedge \bigwedge_{k=1, \dots, n} \theta_k(x))$, a contradiction.

By compactness, we can find a model M of $\mathcal{T}$ such that for some $c \in M^x$ we have $M \models p(c) \wedge \exists y \phi(c, y)$. Let M' be an $|N|^+$ -saturated elementary extension of M . Then one can extend $a \mapsto c$ to a morphism from $N \rightarrow M'$. Since $M' \models \exists y \phi(c, y)$ and N is ec (in the new language, therefore in the original) we get $N \models \exists y \phi(a, y)$.

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